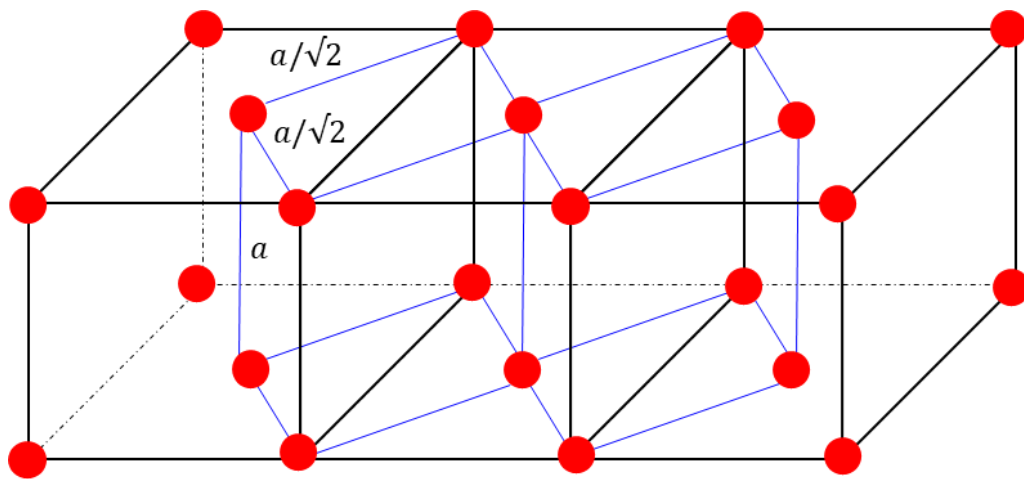


SOLUTIONS TO MID SEM QUESTION PAPER 2025

1. Demonstrate with clear diagram that the (i) **cubic C** and **tetragonal P**, as well as the (ii) **cubic F** and **tetragonal I**, unit cells can generate the same lattice structure. Additionally, explain why the **cubic C** Bravais lattice does not exist, whereas both **cubic F** and **tetragonal I** are valid unit cells. 2+2+2= CO-1
6

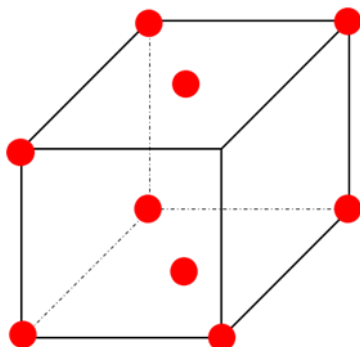
Solution

- (i) **cubic C** and **tetragonal P** unit cells can generate the same lattice structure



$$a = b = c$$

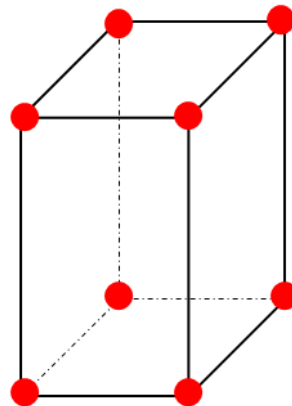
$$\alpha = \beta = \gamma = 90^\circ$$



Cubic C

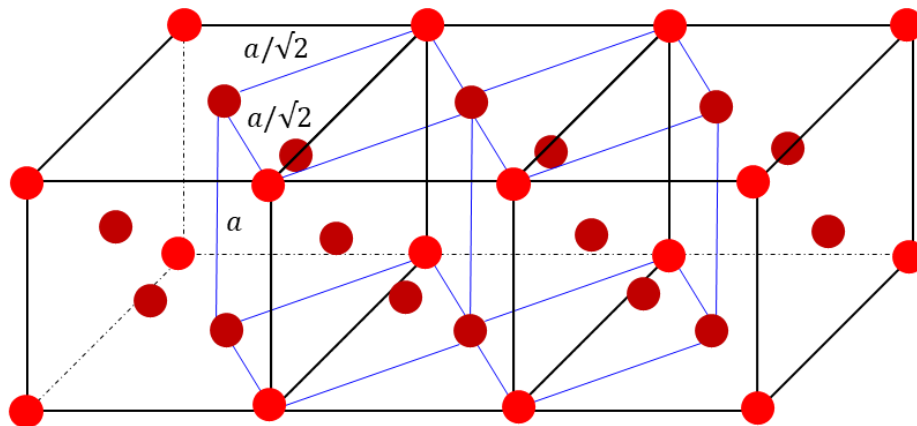
$$a = b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$



Tetragonal P

(ii) **cubic F** and **tetragonal I**, unit cells can generate the same lattice structure

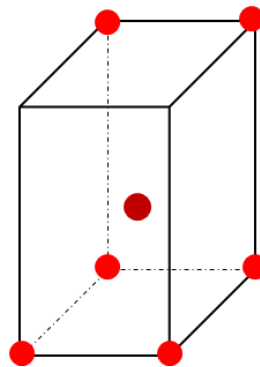
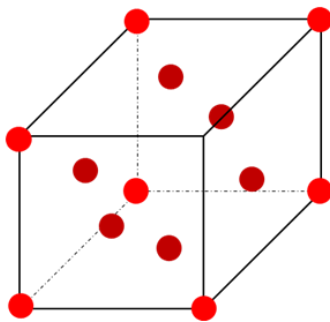


$$a = b = c$$

$$\alpha = \beta = \gamma = 90^\circ$$

$$a = b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$



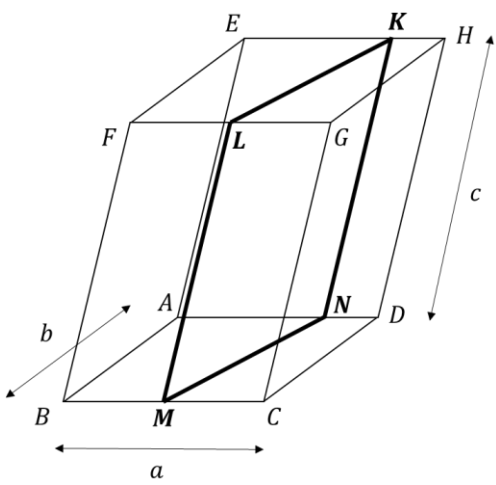
Cubic C Bravais lattice doesnot exist due to the following reasons

1. Lack of Translational Symmetry

A C-centered lattice means there is an additional lattice point at the center of each face perpendicular to the C-axis. However, in a cubic system, all three axes should be equivalent. This additional centering breaks the translational symmetry required for a cubic Bravais lattice.

2. Violation of Cubic Symmetry (Four 3-Fold Rotation Axes)

A cubic crystal system must have four 3-fold rotation axes along the body diagonals. The C-centered cubic arrangement disrupts this requirement, meaning it does not retain the full cubic symmetry. This lack of four 3-fold symmetry disqualifies it from being a Bravais lattice in the cubic system.

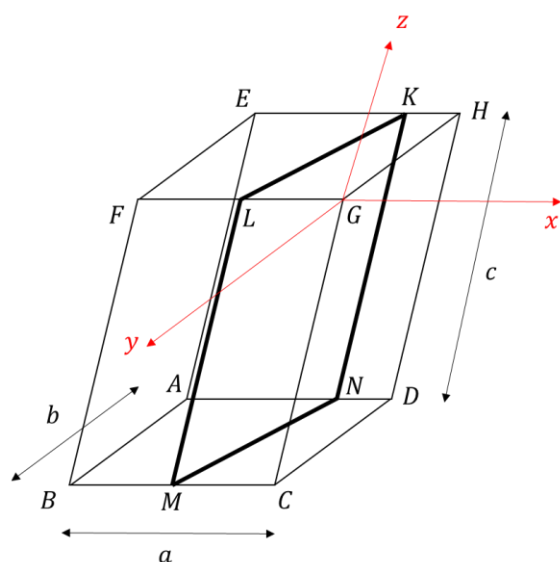
2.	What is the significance of miller indices in crystallography? Calculate the miller indices of the plane KLMN in the figure given below. 	1+5=6	CO-1
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Solution

First Part:

Miller indices are significant in crystallography as they denote both **crystal directions** [uvw] and **crystal planes** (hkl). Miller indices is a standard method to identify or designate crystallographic direction and planes.

Second Part:

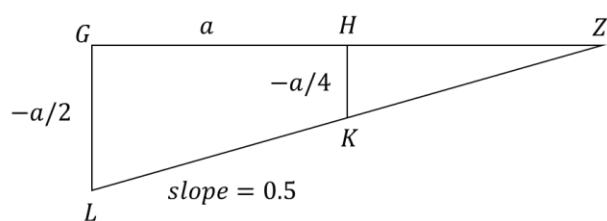


$$a = b, \quad FL = \frac{a}{2}, \quad AN = \frac{3b}{4}$$

$$\text{Intercept at x axis} = -GL = -a/2 = -1/2 \cdot a$$

$$\text{Intercept at Z axis} = \alpha$$

Intercept at y axis



$$GZ = 2a \quad \text{Intercept at Y axis} = 2a$$

$$\text{Intercept at x axis} = -GL = -a/2 = -1/2 \cdot a$$

$$\text{Intercept at Y axis} = 2a = 2 \cdot b$$

$$\text{Intercept at Z axis} = \alpha = \alpha \cdot c$$

$$\text{Reciprocals: } (-2, 1/2, 0)$$

$$\text{Normalized: } (\bar{4}, 1, 0)$$

3.	What is Bragg's Law, and how does it explain X-ray diffraction (XRD) in crystals? In an XRD experiment, a crystal produces a first-order diffraction peak at an angle 40°. Due to applied stress, the interplanar spacing (d) decreases by 20%. (a) What was the original d if the X-ray wavelength used is 1.54 Å? (b) After the 20% reduction, at what diffraction angle will the peak appear?	3+3=6	CO -1
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Solution

First Part:

Bragg's Observation:

When X-rays are incident on a crystal, they interact with atomic planes. The diffracted beams appear as if they are specularly reflected from a set of parallel lattice planes. This reflection-like behavior occurs due to constructive interference between scattered waves.

Bragg's Law:

To explain this observation mathematically, Bragg's Law was formulated as:

$$n\lambda = 2d\sin\theta$$

Where n is the order of diffraction, λ is the wavelength of X-ray, d is the interplanar spacing and θ is the Bragg's angle.

Explanation of X-ray Diffraction (XRD) in Crystals:

X-rays interact with a crystal, and atoms act as scattering centers. When the path difference between waves reflected from successive planes is an integer multiple of the wavelength ($n\lambda$), constructive interference occurs, producing a diffracted beam. By scanning different angles θ , an XRD pattern is obtained, revealing structural details of the crystal.

Second Part:

Given that,

order of diffraction, $n = 1$

Diffraction angle, $2\theta = 40^\circ$

$$\theta = 20^\circ$$

(a) Given, $\lambda = 1.54 \text{ \AA} = 1.54 * 10^{-10} \text{ m}$

We know from Bragg's law,

$$n\lambda = 2d\sin\theta$$

$$\begin{aligned} d &= \frac{n\lambda}{2\sin\theta} \\ &= \frac{1 * 1.54 * 10^{-10}}{2\sin 20} \\ &= \frac{1 * 1.54 * 10^{-10}}{2 * 0.342} \\ &= 2.25 * 10^{-10} \end{aligned}$$

So the original d or interplanar spacing of the crystal is **2.25 Å**

(b)

If d reduces by 20%,

$$d_n = d - 0.2d = 0.8d = 1.8 \text{ \AA}$$

We know from Bragg's law,

$$n\lambda = 2d\sin\theta$$

$$\theta = \sin^{-1}\left(\frac{n\lambda}{2d}\right)$$

$$\theta_n = \sin^{-1}\left(\frac{n\lambda}{2d_n}\right)$$

$$\theta_n = \sin^{-1}\left(\frac{1 * 1.54 * 10^{-10}}{2 * 1.8 * 10^{-10}}\right) = \sin^{-1}(0.427)$$

$$\theta_n = 25.27^\circ$$

$$2\theta_n = 50.54^\circ$$

So the diffraction peak will appear at an angle **50.54°**

4.	Derive the expression for electron energy based on Bohr's model and show how it explained the discrete nature of emission/absorption spectra of hydrogen atom.	4+2=6	CO-2
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Solution

Bohr's model of atom is based on the idea that electrons move in fixed circular orbits around the nucleus with quantized energy levels.

For an electron of charge e and mass m , moving in a circular orbit around a nucleus of charge $+Ze$ (for hydrogen, $Z=1$), the centripetal force required to maintain circular motion is provided by the electrostatic Coulomb force.

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze(-e)}{r^2} = -\frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2}{r^2}$$

The negative direction indicates that the force is attraction.

$$\frac{mv_n^2}{r_n} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2}{r^2}$$

For hydrogen atom, $Z=1$

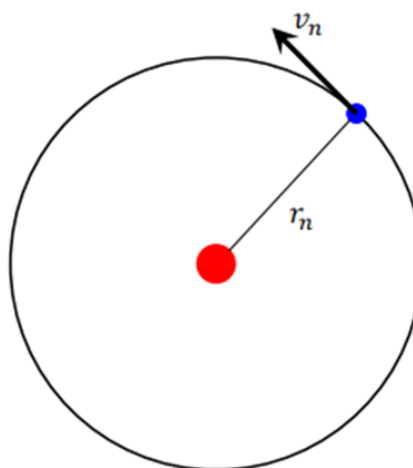
$$\frac{mv_n^2}{r_n} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2}{r_n^2}$$

$$mv_n^2 r_n = \frac{e^2}{4\pi\epsilon_0} \rightarrow (1)$$

Also, as per Bohr's postulate

$$L_n = mv_n r_n = \frac{nh}{2\pi}$$

$$mv_n r_n = n\hbar \rightarrow (2) \quad \hbar = \frac{h}{2\pi}$$



Dividing (1) by (2), we get,

$$v_n = \frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{\hbar} \cdot \frac{1}{n} \rightarrow (3)$$

$$r_n = \frac{n\hbar}{mv_n} = \frac{n\hbar}{m} \cdot \frac{4\pi\epsilon_0 n\hbar}{e^2}$$

$$r_n = 4\pi\epsilon_0 \frac{n^2 \hbar^2}{me^2} \rightarrow (4)$$

Energy,

$$E_n = K_n (K.E.) + U_n (P.E.)$$

$$K_n = \frac{1}{2}mv_n^2 = \frac{1}{2}m \left(\frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{\hbar} \cdot \frac{1}{n} \right)^2$$

$$K_n = \frac{1}{32\pi^2\epsilon_0^2} \cdot \frac{me^4}{\hbar^2} \cdot \frac{1}{n^2}$$

$$U_n = \frac{1}{4\pi\epsilon_0} \cdot \frac{-e^2}{r_n}$$

$$U_n = -\frac{1}{16\pi^2\epsilon_0^2} \cdot \frac{me^4}{\hbar^2} \cdot \frac{1}{n^2}$$

$$E_n = \frac{1}{32\pi^2\epsilon_0^2} \cdot \frac{me^4}{\hbar^2} \cdot \frac{1}{n^2} - \frac{1}{16\pi^2\epsilon_0^2} \cdot \frac{me^4}{\hbar^2} \cdot \frac{1}{n^2}$$

$$E_n = -\frac{1}{32\pi^2\epsilon_0^2} \cdot \frac{me^4}{\hbar^2} \cdot \frac{1}{n^2} = -E_0 \frac{1}{n^2} \quad E_0 = 13.6 \text{ eV}$$

Explanation of Discrete Spectra of Hydrogen Atom

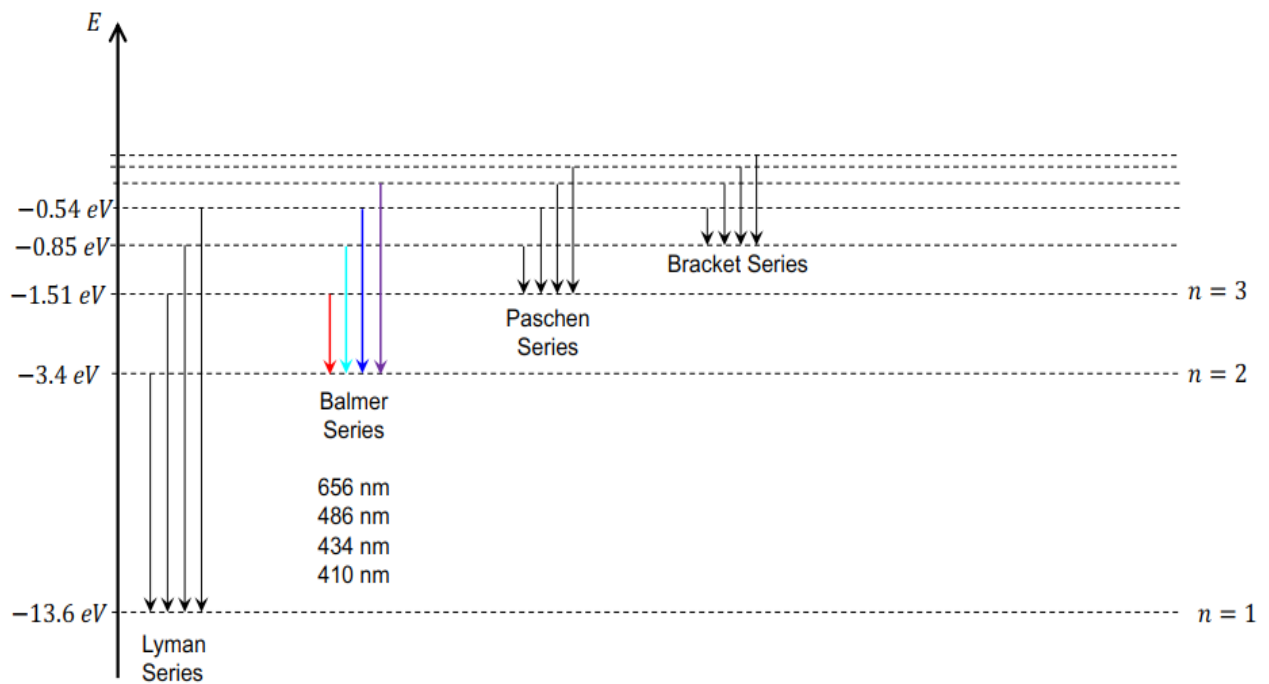
The energy of an electron in a hydrogen atom is restricted to discrete values given by:

$$E_n = -\frac{13.6}{n^2} \text{ eV}$$

When an electron absorbs a photon of energy $E = h\nu$, it jumps to a higher energy level. On the other hand when an electron falls to a lower energy level, it emits a photon of energy $h\nu = E_i - E_f$

Since E depends on n , the emitted photons have specific wavelengths, forming discrete spectral lines.

Spectral Series: Different transitions correspond to different spectral series:



5.	Explain briefly about the wave particle duality. An electron is accelerated through a potential difference of 54.0 V. Calculate the wavelength of the electron using de-Broglie equation. Given that the mass of the electron is 9.1×10^{-31} Kg.	2+4=6	CO-2
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Solution

Wave-Particle Duality: It is the concept in quantum mechanics that every particle exhibits both wave-like and particle-like properties. For example, electrons show diffraction (wave nature) as well as localized impacts in detectors (particle nature).

De Broglie Hypothesis: In 1924, Louis de Broglie proposed a new speculative hypothesis that electrons and other particles of matter can behave like waves.

Particles, like electrons, have an associated wavelength given by $\lambda = h/p$, where h is Planck's constant and p is the particle's momentum.

$$\text{wavelength, } \lambda = \frac{h}{p}$$

$$\text{Energy of electron} \quad qV = 54 \text{ eV} = 54 \times 1.6 \times 10^{-19} \text{ J}$$

$$\text{Also, KE, } E = \frac{1}{2}mv^2 \Rightarrow mE = \frac{1}{2}(mv)^2$$

$$2mE = p^2$$

$$p = \sqrt{2mE} = \sqrt{2 \times 9.1 \times 10^{-31} \times 54 \times 1.6 \times 10^{-19}} = 3.965 \times 10^{-24}$$

$$\text{wavelength, } \lambda = \frac{6.626 \times 10^{-34}}{3.965 \times 10^{-24}} = 0.167 \text{ nm}$$